

Management of Heterogeneous Data in Digital Editions: A Model-Driven Approach

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Abstract

The increasing use of web technologies in the Digital Humanities presents new opportunities for handling domain-specific content but also introduces challenges, particularly due to the heterogeneity of data models. Users of this domain often encounter difficulties in finding or configuring systems for their Digital Editions that can manage complex documents and perform necessary adjustments. This frequently results in self-management, where users must set up and maintain intricate technical infrastructures. To address these issues, a multi-level modeling architecture is proposed for integrating and managing heterogeneous data in Digital Editions. This model-driven infrastructure abstracts elements and relationships across various projects, enhancing interoperability and customization. Key contributions include the development of a meta-model, the implementation of multi-level modeling techniques, and the creation of tools such as a modeling tool, data importer, text editor, and web page generator. These tools are going to improve the efficiency, flexibility, and sustainability of essential processes in Digital Editions, encompassing the conception, elaboration, and publication phases. The approach is going to evaluate, how a model-driven infrastructure based on a platform-independent multi-level modeling approach will improve the efficiency and applicability of the technology in the aforementioned steps of Digital Editions.

CCS Concepts

• **Information systems** → *Web applications; Data model extensions; Graph-based database models*; • **Applied computing** → **Arts and humanities**; • **Software and its engineering** → **Software notations and tools; Model-driven software engineering**.

Keywords

Digital Humanities, Web Technology, Data Model, Model-Driven Development (MDD), Meta-Modeling, Multi-Level Modeling

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1 Problem

The increasing use of web technologies in the Digital Humanities offers new opportunities for working with domain-specific content but simultaneously introduces new challenges. In particular, the heterogeneity of data models in Digital Humanities projects contrasts with the development, maintenance, and extension of infrastructure for Digital Editions [41] [35]. Scholars often face significant difficulties in finding or configuring systems for their Digital Editions that can handle complex documents and perform necessary adjustments. Consequently, many are forced into self-managing, which involves setting up and maintaining intricate technical infrastructures [35]. Existing tools like Codex [34] rely on outdated technology and lack the necessary modernization to ensure compatibility. Many of these tools have not been updated in years, which raises concerns regarding their security and reliability [41].

Digital Editions are scholarly publications that provide foundational material for research in the humanities and cultural studies in digital form [36]. The essential process of their creation includes the **conception**, **elaboration**, and **publication** [16] [36]. During the conception phase, the data model is developed, the subject of the edition is determined, contents and questions are defined, and the relationships within and outside the edition are established. Elaboration involves exploring the contents through the description and transcription of documents, as well as establishing metadata standards. Publication includes the release of the edited subjects in a suitable environment [36]. A concrete example of a Digital Edition is *The Socinian Correspondence* [8]. This project aims to identify letters where at least one correspondent was a Socinian and compile them into a Digital Edition [40]. During the conception phase, a data model was developed and later used in the elaboration phase to describe the correspondences. The resulting edition data was published on a website and made accessible to the public [40].

Currently, there are numerous architectures and tools for the various aforementioned steps of creating Digital Editions, based on diverse approaches and standards [36]. However, these often require cumbersome model transformations of the actual data model of the edition. Further, the variety and complexity of technologies in the Digital Humanities often lead to inadequate and inefficient tools that do not meet specific research requirements [41]. Additionally, these tools often lack user-friendliness and demand significant dedication to master [35]. Large, standardized infrastructures are often too rigid, costly, and unable to adequately support the dynamic, innovative needs of Humanities research [41]. These issues underscore the problems of **heterogeneous data**, **obsolete tools** and **complex technical infrastructures**, forcing scholars to manage intricate setups for Digital Editions due to a lack of user-friendly tools, leading to time-consuming self-management.

In model-driven development (MDD), models are fundamental and serve as primary artefacts in the software development process [2] [29]. In this context, domain-specific languages (DSL) can be used to extend software models with domain-oriented aspects [21] [29]. Using multi-level modeling, further model layers can be created based on a meta-model, which can extend the existing model with additional conditions and aspects [19]. These practices can also be applied to heterogeneous data models of Digital Humanities, enabling the creation of more flexible and adaptable infrastructures, addressing the challenges of interoperability and integration.

Thus, the meta-model presents the abstraction of elements and relations from data models of existing Digital Editions. Based on this, a domain-specific layer can be created, expanding the foundation with specifications of various projects [31]. Even with the meta-model alone, a variety of software solutions can be covered, exemplifying how data sets can be incorporated or Digital Editions generated. By expanding to multiple model layers, customized platforms can be developed additionally.

2 Expected contributions

The overall objective is to develop a model-driven infrastructure that includes an additional domain-specific model layer based on a meta-model. This infrastructure will enable the integration and management of heterogeneous data for Digital Editions in the Digital Humanities. In particular:

- Enhancing the interoperability of various elements within Digital Editions by providing a common meta-model that can be adapted to different projects. For instance, a meta-model could standardize tools that streamline workflows in Digital Editions making it easier and faster for scholars to access and manage different stages of their projects.
- Allow domain experts to tailor the domain-specific layer and tools to fit the specific needs of their projects, ensuring that unique aspects of their data are adequately addressed. A literature scholar might configure tools to handle thematic annotations and inter-textual references, to better elaborate and present their research.
- Facilitate the reuse of developed tools and methodologies across multiple projects, ensuring sustainability and reducing duplication of effort. For example, a tool developed for annotating medieval manuscripts can be easily adapted and reused for modern literary archives, enhancing productivity and promoting consistency across various projects.

3 Related work

Approaches from existing model-driven development projects in the Digital Humanities for Digital Editions illustrate the diversity and potential of model-driven development, but they address different problems with specific, tailored software solutions [41]. The technologies and models used still result in heterogeneity, complicating the interoperability and re-usability of solutions. For instance:

- The Digital Edition of DER STURM [37] converts project-specific XML into static HTML based on concepts from flat-file CMS systems like Jekyll [20], Hugo [18], and Grav [17]. The technological focus is on the sustainability and

accessibility of Digital Editions, ensuring interoperability by generating data formats within the specified data model.

- The CensusIRL project [9] demonstrates the use of XMDD [26], which is utilized in the graphical low-code modeling environment DIME [1] and is used for model-driven development of web applications. While the specific attributes and models are tailored to the needs of the CensusIRL project, they are based on generic principles of data modeling and process definition that could be reused in other similar projects.
- The text editor Codex focuses on complex, overlapping text annotations and is based on its own underlying abstract graph meta-model [34]. It is an independent application that requires model transformations for the integration of other technologies.

These projects already highlight the need for solutions that address the sustainability, accessibility, and interoperability of Digital Editions. However, the current lack of a cohesive framework to manage heterogeneous data models hinders the efficiency and applicability of technologies. This gap results in duplicated efforts, with multiple tools being created based on different technologies and data models, but aiming for the same goals [35] [3]. It also leads to inconsistent data handling, with disparate methods for managing content in Digital Editions [25], and limited scalability across different projects due to the restricted applicability of existing tools to individual projects [41] [36]. These issues underscore the necessity for a unified approach to manage heterogeneous data models.

Platform-dependent data models come in various forms but typically utilize similar data. Simplifying their development and maintenance can be achieved by abstracting the differences in data representation and creating a platform-independent model [31]. The development of a platform-independent meta-model is not a new approach to solving heterogeneous problems. The NIEM specification [33] already describes a platform-dependent and platform-independent meta-model as well as their model transformations [31]. Furthermore, transitioning from a platform-independent model to a multi-level modeling approach can enhance flexibility and scalability by providing deeper instance characterization. This multi-level methodology allows for a more intuitive representation of domain-specific concepts [19]. Moreover, this multi-level approach can bridge the interoperability gap between different tools, enhance scalability across projects, and improve data handling efficiency.

4 Proposed approach

The proposed approach is the development of a model-driven infrastructure for Digital Editions based on a platform-independent meta-model. This meta-model abstracts elements and relationships from heterogeneous data models in existing Digital Editions, transforming them into syntax-neutral units that are independent of the underlying technologies [31] [10]. Building on this, various projects can derive their own domain-specific data models using the type-object pattern [28] [19]. The advantage here lies in the interoperability and re-usability of various elements of the model in heterogeneous data structures [31]. An exemplary abstraction of heterogeneous data models is demonstrated in Figure 1. The Figure shows the transformation of specific graph data models

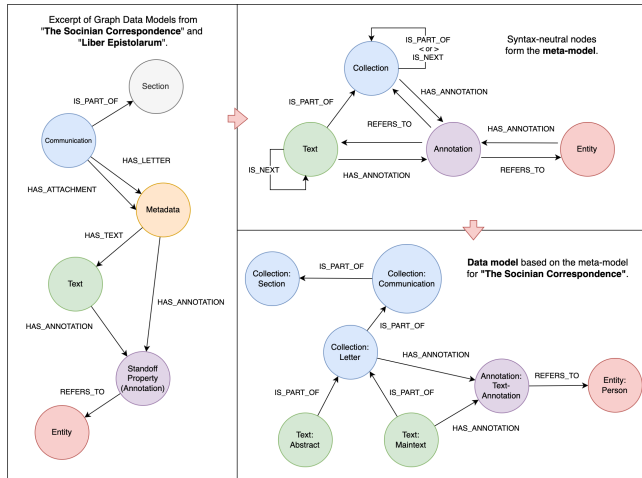


Figure 1: Example abstraction to a meta-model and subsequent transformation into a domain-specific data model.

from two existing Digital Editions *The Socinian Correspondence* [8] and *Liber Epistolarum* [45] into a more abstract meta-model. This meta-model then serves as a basis for deriving domain-specific data models, illustrated here by the data model for *The Socinian Correspondence*. The approach showcases the flexibility and re-usability of the meta-model, allowing diverse projects to create interoperable and consistent data structures, promoting better data integration and analysis across heterogeneous sources.

The use of multi-level modeling [19] enables the implementation of uniform or customized software solutions within the target infrastructure shown in Figure 2 [32] [38]. This expands the range of applicable technologies and increases the efficiency of individual operations within the target infrastructure. Furthermore, the connection of additional developers and projects through the meta-model in multi-level modeling can be improved. To demonstrate the possibilities, a text editor for elaborating and capturing heterogeneous edition data will be developed based on the meta-model [23].

Moreover, data importers need to be developed for existing projects in the Digital Humanities to support model transformation of existing data models and their data into multi-level modeling [41]. Data storage can be managed using the graph database Neo4j, as it can label objects multiple times and thus integrates well with the type-object pattern [6]. Finally, the publication of edition data should be enabled using a website generator. Exemplary tools from the Epsilon family can be used to develop model transformations and code generators [4]. By generating static pages and potentially providing research data on open-source distribution platforms, the sustainability and accessibility of Digital Editions can be enhanced.

For modeling the domain-specific data model during the conception phase, experts from edition projects will be provided with a tool. For its implementation, a web-based environment is planned. Desktop-based graphical frameworks such as GMF [12], Eugenia [22], Graphiti [44], or Sirius [42] are not considered. Web-based frameworks such as WebGME [27], AToMPM [5], SiriusWeb [15], and EMF.Cloud [13] either lack the capability for domain-oriented development [39] or they include components that are not required

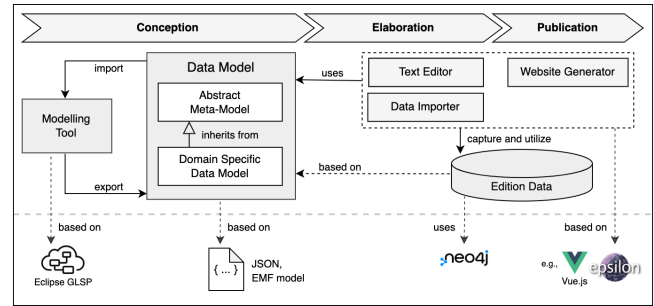


Figure 2: Model-driven target infrastructure.

for the implementation. In contrast, Eclipse GLSP [14] is a modular, extensible, and cross-platform open-source framework that is increasingly being used in research [30]. Eclipse GLSP uses a serialized graphical model for the visual representation of the source model within the framework. The format of the source model is oriented towards the meta-model, implemented in a domain-oriented manner [39] [30], and will be used for model transformations between the graphical and source models [14]. Predefined base models such as EMF, EMF.Cloud, or the graphical model itself can also be used for the source model [30]. The preference for graphical modeling arises from preferences of users in the Digital Humanities [11].

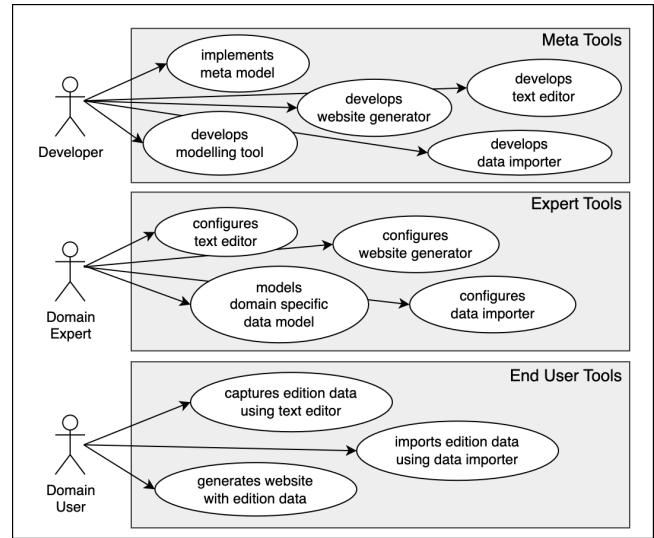


Figure 3: Use cases for the target infrastructure.

Essential interactions and dependencies of actors within the target infrastructure to be developed are visualized in the use case diagram in Figure 3. In particular:

- In the context of this work, **developers** are responsible for implementing the meta-model, the modeling tool, and specific software solutions, such as a website generator, text editor, and data importer. Developers fundamentally extend the model-driven infrastructure with software solutions and contribute to improving the workflow.

- **Domain experts** are experts of a project who have knowledge of the specifications and the possible data model of the project. They use the specialized modeling tool to create and adapt domain-specific data models based on the meta-model. Additionally, they configure the provided tools.
- **Domain users** are project staff members. They use the developed software solutions to capture or import project data and publish them as Digital Editions. Specifically, they capture edition data with the text editor, import edition data with the data importer, and generate websites.

There are no risks or ethical issues associated with this research.

5 Plan for evaluation

Based on the goal of developing a model-driven infrastructure for Digital Editions the central research question arises:

How can a model-driven infrastructure based on a platform-independent multi-level modeling approach be developed to manage the heterogeneity of data models in the Digital Humanities and improve the efficiency and applicability of the technology in the aforementioned steps of Digital Editions?

To answer this question, various methodological approaches are pursued. Usability and functionality tests of the developed tools, such as the text editor, data importer, and website generator, evaluate their functionality, user-friendliness, and efficiency. Additionally, user surveys using questionnaires are conducted to capture the satisfaction and acceptance of users and experts in real Digital Humanities projects. A technical evaluation of the performance of the new infrastructure in these projects rounds off the investigation.

The evaluation of the target infrastructure includes several key components, including the multi-level approach and the tools for conceiving, editing and publishing Digital Editions. The assessment of the tools used in the various workflow steps will examine the functionality, usability, and efficiency of the text editor, the data importer, and the website generator. Usability tests with domain users, functionality tests to check technical performance, and questionnaires to capture user satisfaction and acceptance form the methodological foundation.

Additionally, questions related to concept, technology, and application will be considered, focusing on the various components of the target infrastructure:

Concept:

- What domain-specific requirements must be considered to develop and deploy the platform-independent multi-level model?
- What challenges arise in deriving domain-specific data models using the type-object pattern?
- What limitations exist in integrating heterogeneous data and their data models?
- How can a domain-specific layer be effectively created and managed based on the meta-model?

Technology:

- How can the implementation on the Eclipse GLSP Framework be advanced agilely, considering the needs of users?

- How can existing data models and their content be transformed into the new multi-level modeling and existing projects be migrated to the model-driven infrastructure?
- How can developers be supported in developing further software solutions in the model-driven infrastructure?

Application:

- What specific advantages and improvements have been achieved through the use of the new tools and the meta-model in real projects?
- To what extent has the range of applicable technologies been expanded and the efficiency of workflows increased?
- What challenges and problems were identified during the application phase, and how were they addressed?

The involvement of domain experts and users, as well as concrete projects and their staff, are essential components of the evaluation. The plan is to implement and evaluate the target infrastructure in the context of the DFG projects:

- **The Book of Letters of Hildegard of Bingen. Genesis - Structure - Composition**, led by Univ.-Prof. Dr. Mechthild Dreyer, Prof. Dr. Thomas Stäcker, and Prof. Dr. Andreas Kuczera [45] [24].
- **Between Theology, Early Modern Natural Science, and Political Correspondence: The Socinian Correspondence**, led by Dr. Kęstutis Daugirdas and Prof. Dr. Andreas Kuczera [8] [7].
- **The Struggle for Integration and Self-Assertion – Edition of Antje Brons' Letters (1810-1902)**, led by Dr. Klaas-Dieter Voß and Prof. Dr. Andreas Kuczera [43].

6 Current status

The recent research focused on identifying the topic and the necessary preliminary work. This includes an initial literature review in the field of platform-independent modeling and the examination of existing Digital Editions in the Digital Humanities. Additionally, it involves identifying open challenges and problems along with their workflows. This has led to a refinement of the scope. Moreover, within the framework of Digital Humanities conferences and workshops, data models of various Digital Editions have been reviewed to create an initial abstraction of these into units that are independent of the underlying technologies. These units have been further refined with the help of domain experts. Nevertheless, further abstraction work is required, and additional data models of other Digital Editions need to be considered. Specifically, the treatment of syntax-neutral text units needs to be reassessed, particularly with regard to other possible objects that can also be annotated or related to other units.

The next steps involve engaging with additional data models of existing editions in Digital Humanities to abstract the meta-model. Initial approaches to domain-specific modeling will be developed based on this meta-model. The process will be component-wise, aligned with the workflow steps of Digital Editions, covering the phases of conception, elaboration, and publication of Digital Editions. This includes using the modeling tool during the conception phase, the text editor and data importer during the elaboration phase, and the website generator for publishing the edition data.

The requirements for the modeling tool will be gathered through user-centered design methods, including defining target groups, conducting domain interviews, questionnaires, focus groups, on-site observations, and use-case scenarios. Agile development following the SCRUM methodology will be implemented, with iterative evaluations involving Digital Humanities stakeholders.

The following components will be developed and assessed to ensure usability and efficiency:

- The modeling tool's usability and efficiency in creating and customizing domain-specific data models will be assessed.
- The text editor will be enhanced to capture edition data based on the domain-specific data model, with evaluations focusing on its efficiency and precision.
- The data importer will be developed to facilitate the migration of existing data models into the new multi-level model, with evaluations measuring its support for seamless data transfer and integration.
- A website generator for static Digital Editions will be developed based on the multi-level model, and its efficiency in creating static web pages for Digital Editions will be evaluated.

In the end, the entire infrastructure will be demonstrated in workshops and conferences, from the modeling tool to distributed software solutions like the text editor, data importer, and website generators.

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